

Steve Gwade

ML Engineer (Applied Research)

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Professional Summary

ML Engineer with 3 years of experience designing, building, and deploying machine learning systems across healthcare, agriculture, finance, transport and consumer products. Skilled in RAG systems, real-time inference pipelines, computer vision systems, unsupervised systems, automation and applied research, with a track record of turning ML ideas into production-ready systems that improve speed, reliability, and decision quality.

Experience

- Freelance Backend Engineer (Applied AI)** Mar 2026 – May 2026
MIT S S A R L – Moabi B2B Travel (Freelance)
 - Engineered a conversational booking workflow using LLM-based intent parsing, bilingual interaction, and approval controls, reducing booking request time by over 90%.
 - Designed decision flows and concurrency-safe approval handling, improving reliability across simultaneous user sessions.
 - Integrated the AI module into an existing event-driven AWS system without changing upstream services, enabling production adoption with minimal platform disruption.
- Research Engineer (ML & Full-Stack)** Jan 2025 – Jan 2026
J I A N T S – Systems Intelligence Lab (Part-time)
 - Conducted heavy internal experimentation with AI tools and workflows for vision, OCR, STT, and TTS use cases, reducing prototype-to-deployment time by about 30%.
 - Built end-to-end PyTorch pipelines for preprocessing, augmentation, multi-task training, and inference optimization.
 - Contributed to LINA, an assistive vision system for blind users, helping improve navigation and object recognition by 56%.
- Software Engineer (AI & Full Stack)** Mar 2024 – Dec 2024
T G N i x – IT Consulting (Part-time)
 - Built and deployed AI-backed systems for client use cases, reduced manual processing and improved operational efficiency.
 - Integrated ML inference pipelines into applications, improving workflow speed by 25% through automated decision support.
 - Developed Python backends and APIs for model serving, system integration, and stable performance under heavy usage.
 - Deployed Dockerized applications to cloud platforms, improving release reliability and reducing deployment-related issues.
- Junior Backend Engineer (ML & Computer Vision)** Jun 2023 – Nov 2023
A I T E C A F – AI Technologies for Africa (Internship)
 - Developed Django REST APIs to serve ML models and non-ML business services efficiently.
 - Built conversational assistants using LangChain and LLMs to automate repetitive client interactions.
 - Trained and evaluated a YOLO-based cassava leaf disease detection model enabling sub-5ms real-time inference.
 - Assisted in data annotation, model validation, and backend testing to ensure production-ready ML deployment.

Selected Projects

- VERO** (*Personal Project, Done*)
 - Built a grounded research workspace that maps documents, URLs, and repos into project-scoped knowledge bases for Q/A.
 - Built a hybrid retrieval pipeline combining dense search, keyword search, fusion, and reranking for more traceable answers.
 - Designed citation-aware answer generation and session memory for reliable multi-turn research over private data.
- PlumVision** (*JCIA Hackathon Winner, 2025*)
 - Built a real-time plum classifier using ConvNeXt and contrastive learning achieving 98% accuracy with a 3MB model.
 - Optimized preprocessing, augmentation, and deployment pipelines to deliver sub-12ms inference latency.
 - Exposed the model via FastAPI for live demos and integration into downstream applications.
- EatWise** (*JCIA Most Innovative AI Prototype Winner, Ongoing*)
 - Developed a food recognition pipeline using object detection and classification for personalized nutrition analysis.
 - Integrated recommendation logic for meal planning based on user habits, goals, and fridge inventory.
 - Served real-time inference via FastAPI backend for mobile-oriented AI workflow.

Other Projects

- Spoof Detector** (*Personal Project, Done*)
 - Trained a deep CNN for face spoof detection to enhance biometric authentication reliability.
 - Exposed model via Flask REST API for integration into authentication workflows.
- Multi-Task Transfer Learning for Malaria Diagnosis** (*MSc Thesis, 2025*)
 - Formulated a multi-task transfer learning objective, enabling joint optimization across heterogeneous tasks.

- Engineered data pipelines for task-specific datasets and align heterogeneous data distributions for multi-task training.
- Analyzed task interactions through experiments, highlighting trade-offs between single-task and multi-task training.
- Achieved a 97% macro F1 score, outperforming single-task baselines and demonstrating robust predictive performance.
- Deployed inference services using FastAPI to support reproducible evaluation and integration.

Technical Skills

- **Programming Languages:** Python, JavaScript, SQL
- **Machine Learning:** Supervised learning, Self-supervised, Unsupervised, Representation learning, Transfer learning
- **Deep Learning Frameworks:** PyTorch, TensorFlow, Keras, Jax
- **Computer Vision:** Image classification, Object detection, Segmentation, Feature extraction
- **Backend & APIs:** RESTful APIs, Django REST Framework, FastAPI, Flask
- **Data & Storage:** PostgreSQL, SQLite, MongoDB
- **Deployment & Tooling:** Docker, CI/CD pipelines, MLOps, GitHub Actions, AWS (EC2, Fargate, S3, SageMaker), Linux, MLflow, Weights & Biases

Awards and Recognition

- **2nd Place ODC Champions Africa 2025**, *Orange Digital Center* – 2nd among all African countries in an algorithmic coding contest on CodinGame with my team.
- **JCIA Hackathon 2025 Winner**, *PlumVision* – developed a real-time plum classifier achieving 98% accuracy, ~3MB model, <12ms inference, enabling instant quality control.
- **Most Innovative AI Prototype JCIA AI Solutions Track Winner**, *EatWise* – mobile AI assistant improving nutritional habits via food recognition, personalized meal plans, and hundreds of local recipes based on users' fridge, habits, and goals.
- **1st Prize Worldwide Huawei ICT Competition (Innovative Track)**, *LINA Assistive Vision System for the Blind* – contributed to building an AI system improving by 56% navigation, autonomy, and object recognition for blind users.

Academic Qualifications

- **MSc in Engineering** (2023–2025), National Higher Polytechnic School of Douala – specialization in Data Science & AI, MScEng thesis: multi-task transfer learning for malaria diagnosis
- **BSc in Engineering** (2020–2023), National Higher Polytechnic School of Douala – focus on Computer Science and foundations of Data Science & AI

Research & Publications

- **Multi-Task Transfer Learning and Application to Malaria Detection** – MSc Thesis, 2025, github.com/GwadeSteve/MTTL-Research-Malaria.

Professional Skills

- **Problem Solving:** Diagnosing and resolving complex ML, data, and system challenges under ambiguous requirements.
- **Strategy & Execution:** Designing experiments, analyzing results, and driving end-to-end ML project delivery.
- **Communication:** Translating technical insights, documenting models, and explaining AI concepts to diverse stakeholders.

Interests

Reading, Gaming, Basketball, Coffee, Open-source contribution